

Web Technology I

Unit 5 - Cascading Style Sheet (CSS)

Unit 5. Cascading Style Sheet (CSS) [5 Hrs.]

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5.1. Introduction to Cascading Style Sheets (CSS) and advantages of using CSS

<https://www.geeksforgeeks.org/css-introduction/>

CSS stands for Cascading Style Sheets. It is the language for describing the presentation of Web pages, including colours, layout, and fonts, thus making our web pages presentable to the users.

Now let's try to break the acronym:

- Cascading: Falling of Styles - (Meaning - arrange)

- **Style:** Adding designs/Styling our HTML tags - (Meaning - design)
- **Sheets:** Writing our style in different documents (Meaning - pages)

CSS handles the look and feel part of a web page. Using CSS, you can control the color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colors are used, layout designs, variations in display for different devices and screen sizes as well as a variety of other effects.

CSS is easy to learn and understand but it provides powerful control over the presentation of an HTML document. Most commonly, CSS is combined with the markup languages HTML or XHTML.

Advantages of CSS (Why CSS?) -

- **CSS saves time** – You can write CSS once and then reuse same sheet in multiple HTML pages. You can define a style for each HTML element and apply it to as many Web pages as you want.
- **Pages load faster** – If you are using CSS, you do not need to write HTML tag attributes every time. Just write one CSS rule of a tag and apply it to all the occurrences of that tag. So less code means faster download times.
- **Easy maintenance** – To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.
- **Superior styles to HTML** – CSS has a much wider array of attributes than HTML, so you can give a far better look to your HTML page in comparison to HTML attributes.
- **Multiple Device Compatibility** – Style sheets allow content to be optimized for more than one type of device. By using the same HTML document, different versions of a website can be presented for handheld devices such as PDAs and cell phones or for printing.
- **Global web standards** – Now HTML attributes are being deprecated and it is being recommended to use CSS. So its a good idea to start using CSS in all the HTML pages to make them compatible to future browsers.
- **Search Engines:** CSS is considered a clean coding technique, which means search engines won't have to struggle to "read" its content.
- **Offline Browsing:** CSS can store web applications locally with the help of an offline cache. Using this we can view offline websites.

Extra –

CSS Comment

Comments don't render on the browser.

Helps to understand our code better and makes it readable.

Helps to debug our code.

Example -

```
/* h1 { color: red; text-align: center; } */
```

Example - HTML and CSS Comments

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red; /* Set text color to red */
}
</style>
</head>
<body>

<h2>My Heading</h2>

<!-- These paragraphs will be red -->
<p>Hello World!</p>
<p>This paragraph is styled with CSS.</p>
<p>CSS comments are not shown in the output.</p>

</body>
</html>
```

5.2. Basic Syntax

5.2.1. Creating Cascading Style Sheets (CSS) using STYLE tag

https://www.tutorialspoint.com/css/css_syntax.htm

<https://www.mygreatlearning.com/blog/css-tutorial/>

CSS Syntax –

Selector

```
{
  Property 1 : value;
  Property 2 : value;
  Property 3 : value;
}
```

Here,

--basic selectors like tags, id's, and classes

--properties(attributes) like color, font-size, background, width, height, etc

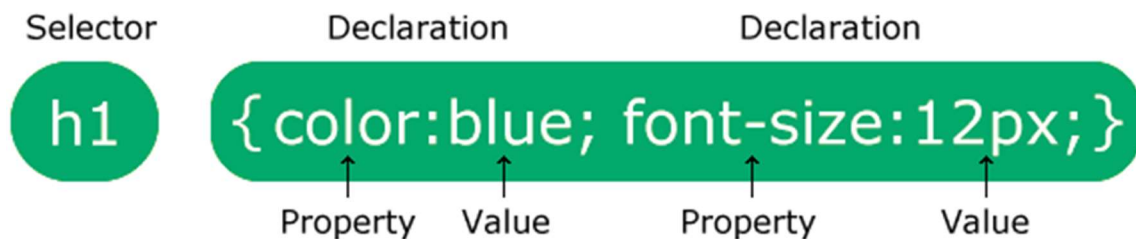
--values associated with these properties.

For example:

```
h1 {  
  Color: red;  
  Text-align: center;  
}  
  
#unique {  
  color: green;  
}
```

CSS Syntax -

A CSS rule consists of a selector and a declaration block.



The selector points to the HTML element you want to style. There are few basic selectors like tags, id's, and classes

The declaration block contains one or more declarations separated by semicolons.

Each declaration includes a CSS property name and a value, separated by a colon.

Multiple CSS declarations are separated with semicolons, and declaration blocks are surrounded by curly braces.

Example –

```
p {  
  color: blue;  
  text-align: center;  
}
```

Hello World!

These paragraphs are styled with CSS.

5.2.1. Creating Cascading Style Sheets (CSS) using STYLE tag

Example 1-

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  color: red;
  text-align: center;
}
</style>
</head>
<body>

<p>Hello World!</p>
<p>These paragraphs are styled with CSS.</p>

</body>
</html>
```

O/P

Hello World!

These paragraphs are styled with CSS.

Example 2 -

```

<!DOCTYPE html>
<html>
<head>
  <title>Example</title>
  <style>
    main {
      width: 600px;
      height: 200px;
      padding: 10px;
      background: beige;
    }

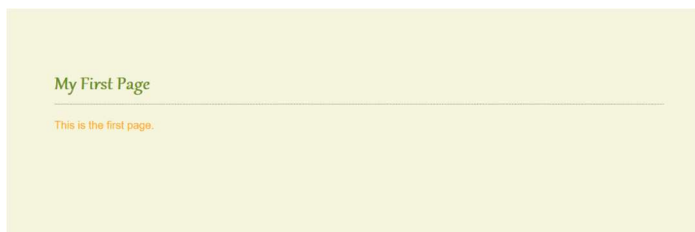
    h1 {
      color: olivedrab;
      border-bottom: 1px dotted darkgreen;
    }

    p {
      font-family: sans-serif;
      color: orange;
    }
  </style>
</head>

<body>
  <main>
    <h1>My first Page</h1>
    <p>This is a basic web page.</p>
  </main>
</body>
</html>

```

Output –



Note –

CSS Selectors

CSS selectors are used to “find” (or select) HTML elements you want to style based on their element name, id, class, attribute, and more. [The selector is used to target elements and apply CSS.]

- Three simple selectors
 - Element Selector
 - Id Selector
 - Class Selector
- Universal Selector

Element Selector –

The element selector selects HTML elements based on the element name.

Example –

```
h1  
{  
  Color: red;  
}
```

We selected the heading tag and then changed the color property i.e text color to red. Now whatever is written in this tag (content) will have the text color as red.

Example -

```
p {  
  text-align: center;  
  color: red;  
}
```

Here, all <p> elements on the page will be center-aligned, with a red text color:

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
p {  
  text-align: center;  
  color: red;  
}  
</style>  
</head>  
<body>  
  
<p>Every paragraph will be affected by the style.</p>  
<p id="para1">Me too!</p>  
<p>And me!</p>  
  
</body>  
</html>
```

O/P

Every paragraph will be affected by the style.

Me too!

And me!

ID Selector –

The id selector uses the id attribute of an HTML element to select a specific element.

The id of an element is unique within a page, so the id selector is used to select one unique element!

To select an element with a specific id, write a hash (#) character, followed by the id of the element.

Example –

```
<h1 id="unique"> Hi </p>
```

CSS will be -

```
#unique
```

```
{
```

```
Color: red;
```

```
}
```

We selected the id and then changed the color property i.e text color to red. Now whatever is written in this tag (content) will have the text color as red .

Example -

The CSS rule below will be applied to the HTML element with id="para1":

```
#para1 {  
  text-align: center;  
  color: red;  
}
```



```

<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
    text-align: center;
    color: red;
}
</style>
</head>
<body>

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by the style.</p>

</body>
</html>

```

O/P

Hello World!

This paragraph is not affected by the style.

Class Selector –

The class selector selects HTML elements with a specific class attribute.

To select elements with a specific class, write a period (.) character, followed by the class name.

Example –

```
<h1 class="group"> Hi </p>
```

CSS will be –

```
.group
```

```
{
```

```
Color: red;
```

```
}
```

We selected the class and then changed the color property i.e text color to red.

Now whatever is written in this tag (content) will have the text color as red.

Example -

In this example all HTML elements with class="center" will be red and center-aligned:

```

.center {
  text-align: center;
  color: red;
}

<!DOCTYPE html>
<html>
<head>
<style>
.center {
  text-align: center;
  color: red;
}
</style>
</head>
<body>

<h1 class="center">Red and center-aligned heading</h1>
<p class="center">Red and center-aligned paragraph.</p>

</body>
</html>

```

O/P

Red and center-aligned heading

Red and center-aligned paragraph.

Universal Selector –

The universal selector (*) selects all HTML elements on the page.

Example –

```

* {
  color: #000000;
}

```

This renders the content of every element in our document in black.

Example –

```

<!DOCTYPE html>
<html>
<head>
<style>
* {
  text-align: center;
  color: blue;
}
</style>
</head>
<body>

<h1>Hello world!</h1>

<p>Every element on the page will be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>

```

O/P

Hello world!

Every element on the page will be affected by the style.

Me too!

And me!

Grouping Selector

The grouping selector selects all the HTML elements with the same style definitions.

Look at the following CSS code (the h1, h2, and p elements have the same style definitions):

```

h1 {
  text-align: center;
  color: red;
}

h2 {
  text-align: center;
  color: red;
}

p {
  text-align: center;
  color: red;
}

```

It will be better to group the selectors, to minimize the code.

To group selectors, separate each selector with a comma.

Example -

In this example we have grouped the selectors from the code above:

```
h1, h2, p {  
  text-align: center;  
  color: red;  
}
```

Extra -

- Priority of Selectors
Id > Class>Element

5.3. Types of Style Sheets

5.3.1. Inline Style Sheets

5.3.2. Internal/ Embedded Style Sheets

5.3.3. External Style Sheets

https://www.w3schools.com/css/css_howto.asp

- There are 3 ways to write CSS in our HTML file.
 - Inline CSS
 - Internal CSS
 - External CSS

Inline CSS

```
<p style="color: blue;">This is a paragraph.</p>
```

Internal CSS

```
<head>  
  <style type = text/css>  
    body {background-color: blue;}  
    p { color: yellow;}  
  </style>  
</head>
```

External CSS

```
<head>  
  <link rel="stylesheet" type="text/css" href="style.css">  
</head>
```

Example –

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
  
  <title>Types of CSS</title>  
  
  <link rel="stylesheet" type="text/css" href="first.css">  
  
<style>  
  
h1  
{  
  color:green;  
}  
  
</style>  
  
</head>  
  
<body>  
  
<h1>This heading will be green</h1>  
  
<p style="color:Red">This paragraph will be red</p>  
  
<p id="center">This paragraph will be pink and center-  
aligned</p>  
  
</body>  
  
</html>
```

```
This is Css file  
first.css  
#center {  
    text-align: center;  
    color: pink;  
}
```

Inline CSS

An inline style may be used to apply a unique style for a single element.

To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

Example

Inline styles are defined within the "style" attribute of the relevant element:

```
<!DOCTYPE html>  
<html>  
<body>  
  
<h1 style="color:blue;text-align:center;">This is a heading</h1>  
<p style="color:red;">This is a paragraph.</p>  
  
</body>  
</html>
```

Internal CSS

An internal style sheet may be used if one single HTML page has a unique style.

The internal style is defined inside the <style> element, inside the head section.

Example

Internal styles are defined within the <style> element, inside the <head> section of an HTML page:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: linen;
}

h1 {
  color: maroon;
  margin-left: 40px;
}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

External CSS

With an external style sheet, you can change the look of an entire website by changing just one file!

Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section.

Example

External styles are defined within the <link> element, inside the <head> section of an HTML page:

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="mystyle.css">
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

An external style sheet can be written in any text editor, and must be saved with a .css extension.

The external .css file should not contain any HTML tags.

Here is how the "mystyle.css" file looks:

"mystyle.css"

```
body {  
  background-color: lightblue;  
}  
  
h1 {  
  color: navy;  
  margin-left: 20px;  
}
```

Extra 1–

- Priority order
 - Inline > Internal > External

Extra 2–

Import CSS Style

<https://www.geeksforgeeks.org/what-is-the-best-way-to-include-css-file-why-use-import/>

Example -

index.html


```

<!DOCTYPE html>
<html>
<head>
  <title>Import CSS Style</title>
  <style>
    @import url("style.css");
  </style>
</head>
<body>
  <p>This is a first paragraph.</p>
  <div>This is a second paragraph.</div>
</body>
</html>

```

style.css

```

/*CSS Style*/
p {
  color:purple;
  margin-left:20px;
}
div{
  color:purple;
  font-size:16px;
  background-color:#FF6633;
}

```

O/P

This is a first paragraph.

This is a second paragraph.

5.4. Introduction to different Styles and their Attributes

5.4.1. Backgrounds and Color Styles and Attributes

CSS Background

<https://www.geeksforgeeks.org/css-background/>

https://www.tutorialspoint.com/css/css_backgrounds.htm

The CSS background properties are used to add background effects for elements.

- **background-color** - Sets the background color of an element.
- **background-image** - Sets the background image for an element.
- **background-repeat** - Sets how a background image will be repeated.
- **background-position** - Sets the starting position of a background image.
- **background-attachment** - Sets whether a background image is fixed or scrolls with the rest of the page.

CSS background-color

The **background-color** property specifies the background color of an element.

Syntax –

```
body {  
    background-color:color-name  
}
```

Example

The background color of a page is set like this:

```
body {  
    background-color: lightblue;  
}
```

With CSS, a color is most often specified by:

- a valid **color name** - like "red"
- a **HEX** value - like "#ff0000"
- an **RGB** value - like "rgb(255,0,0)"

Example

Here, the <h1>, <p>, and <div> elements will have different background colors:

```
h1 {  
    background-color: green;  
}  
  
div {  
    background-color: lightblue;  
}  
  
p {  
    background-color: yellow;  
}
```

Example -

```
<!DOCTYPE html>
<html>
<head>
  <style>
    h1 {
      background-color: blue;
    }
  </style>
</head>

<body>
  <h1>Geeksforgeeks</h1>
</body>
</html>
```

O/P



Extra -

Opacity / Transparency

The **opacity** property specifies the opacity/transparency of an element. It can take a value from 0.0 - 1.0. The lower value, the more transparent:

Example

```
div {
  background-color: green;
  opacity: 0.3;
}
```

Example

```
div {
  background: rgba(0, 128, 0, 0.3) /* Green background with 30% opacity
*/
}
```

CSS background-image

The `background-image` property specifies an image to use as the background of an element. By default, the image is repeated so it covers the entire element.

Syntax –

```
body {  
    background-image : link;  
}
```

Example

Set the background image for a page:

```
body {  
    background-image: url("paper.gif");  
}
```

The background image can also be set for specific elements, like the `<p>` element:

Example

```
p {  
    background-image: url("paper.gif");  
}
```

Example –

```
<html>  
<head>  
  <style>  
    body {  
      background-image: url("/css/images/css.jpg");  
      background-color: #cccccc;  
    }  
  </style>  
</head>  
  
<body>  
  <h1>Hello World!</h1>  
</body>  
</html>
```

CSS background-repeat

By default, the `background-image` property repeats an image both horizontally and vertically.

Some images should be repeated only horizontally or vertically, or they will look strange, like this:

Syntax –

To repeat an image horizontally

```
body {  
    background-image: link;  
    background-repeat: repeat-x;  
}
```

To repeat an image Vertically

```
body {  
    background-image: link;  
    background-repeat: repeat-y;  
}
```

Example

```
body {  
    background-image: url("gradient_bg.png");  
    background-repeat: repeat-x;  
}
```

Example –

```
<html>  
<head>  
  <style>  
    body {  
      background-image: url("/css/images/css.jpg");  
      background-repeat: repeat; | repeat-x; | repeat-y;  
    }  
  </style>  
</head>  
  
<body>  
  <p>Tutorials point</p>  
</body>  
</html>
```

Extra –

CSS background-repeat: no-repeat

Showing the background image only once is also specified by the `background-repeat` property:

Example

Show the background image only once:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
}
```

CSS background-position

The `background-position` property is used to specify the position of the background image.

Example

Position the background image in the top-right corner:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
}
```

Example –

```
<html>  
<head>  
  <style>  
    body {  
      background-image: url("/css/images/css.jpg");  
      background-position: 100px 200px;  
    }  
  </style>  
</head>  
  
<body>  
  <p>Tutorials point</p>  
</body>  
</html>
```

This example demonstrates how to set the background image position 100 pixels away from the left side and 200 pixels down from the top.

CSS background-attachment -

The `background-attachment` property specifies whether the background image should scroll or be fixed (will not scroll with the rest of the page):

Example

Specify that the background image should be fixed:

```
body {  
  background-image: url("img_tree.png");  
  background-repeat: no-repeat;  
  background-position: right top;  
  background-attachment: fixed;  
}
```

CSS Colors -

<https://www.geeksforgeeks.org/css-colors/>

https://www.tutorialspoint.com/css/css_colors.htm

CSS Color property is used to set the color of HTML elements. This property is used to set font color, background color etc.

Color of an element can be defined in the following ways:

- **Built-In Color** → Color name, eg – red, green etc.
- **RGB Format** → rgb(red, green, blue), eg- rgb(255, 99, 71)
- **RGBA Format** → rgba(R, G, B, A) A- Alfa for transparency, Eg - rgba(0, 153, 0, 0.5)
- **Hexadecimal Notation** - > Red - #FF0000 , Green- #00FF00
- **HSL** → Hue, Saturation, and Lightness - hsl(120, 100%, 30%)
- **HSLA** → hsla(120, 100%, 50%, 0.50)

CSS Background Color -

You can set the background color for HTML elements.

Example

```
<h1 style="background-color:DodgerBlue;">Hello World</h1>  
<p style="background-color:Tomato;">Lorem ipsum...</p>
```

CSS Text Color

You can set the color of text.

Example

```
<h1 style="color:Tomato;">Hello World</h1>  
<p style="color:DodgerBlue;">Lorem ipsum...</p>  
<p style="color:MediumSeaGreen;">Ut wisi enim...</p>
```

Hello World

Lorem ipsum dolor sit
dolore magna aliquam

Ut wisi enim ad minim
commodo consequat.

CSS Border Color

You can set the color of borders:

Example

```
<h1 style="border:2px solid Tomato;">Hello World</h1>  
<h1 style="border:2px solid DodgerBlue;">Hello World</h1>  
<h1 style="border:2px solid Violet;">Hello World</h1>
```

Hello World

Hello World

Hello World

5.4.2. Fonts and Text Styles and Attributes

CSS Fonts –

<https://www.geeksforgeeks.org/css-fonts/>

The CSS font is used to set the font's content of the HTML element.

You can set following font properties of an element –

- The **font-family** property is used to change the face of a font.
- The **font-style** property is used to make a font italic or oblique.
Possible values are *normal*, *italic* and *oblique*.
- The **font-variant** property is used to create a small-caps effect.
Possible values are *normal* and *small-caps*.
- The **font-weight** property is used to increase or decrease how bold or light a font appears.
Possible values could be *normal*, *bold*, *bolder*, *lighter*, *100*, *200*, *300*, *400*, *500*, *600*, *700*, *800*, *900*.

- The **font-size** property is used to increase or decrease the size of a font. Possible values could be *xx-small*, *x-small*, *small*, *medium*, *large*, *x-large*, *xx-large*, *smaller*, *larger*, *size in pixels or in %*.

Example –

Ex –

font-family -

font-family: "Times New Roman", Times, serif;

font-family: Arial, Helvetica, sans-serif;

font-family: "Lucida Console", "Courier New", monospace;

font-style -

font-style: normal;

font-style: oblique;

font-style: italic;

font-variant –

font-variant: normal;

font-variant: small-caps;

font-weight -

font-weight: normal;

font-weight: bold;

font-weight: 500;

font-size -

font-size: 40px; | **font-size**: 2.5em; /* 40px/16=2.5em */

font-size: small;

font-size: large;

Example –

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>font-size property</title>
```

```
  <style>
```

```
    .gfg {
```

```

        font-size: 40px;
        font-weight: bold;
        font-family: "Times New Roman";
        color: #090;
        text-align: center;
    }

    .geeks {
        font-size: 1.2em;
        text-align: center;
    }
</style>
</head>

<body>
    <div class="gfg">GeeksforGeeks</div>
    <div class="geeks">
        A computer science portal for geeks
    </div>
</body>

</html>

```

CSS Text –

<https://www.geeksforgeeks.org/css-text-formatting/>

https://www.tutorialspoint.com/css/css_text.htm

The CSS **text formatting** properties are used to format text, style the text and perform different types of manipulations like word spacing, alignment, justification, and text transformation.

Syntax –

```

Selector {
property-name : /*value*/
}

```

You can set following text properties of an element –

- The **color** property is used to set the color of a text.
Possible value could be any color name in any valid format.
- The **direction** property is used to set the text direction.
Possible values are *ltr* or *rtl*.
- The **letter-spacing** property is used to add or subtract space between the letters that make up a word.

Possible values are *normal* or *a number specifying space*.

- The **word-spacing** property is used to add or subtract space between the words of a sentence.

Possible values are *normal* or *a number specifying space*.

- The **text-indent** property is used to indent the text of a paragraph.

Possible values are *%* or *a number specifying indent space*.

- The **text-align** property is used to align the text of a document.

Possible values are *left*, *right*, *center*, *justify*.

- The **text-decoration** property is used to underline, overline, and strikethrough text.

Possible values are *none*, *underline*, *overline*, *line-through*, *blink*.

https://www.w3schools.com/css/css_text_decoration.asp

- The **text-transform** property is used to capitalize text or convert text to uppercase or lowercase letters.

Possible values are *none*, *capitalize*, *uppercase*, *lowercase*.

- The **white-space** property is used to control the flow and formatting of text.

Possible values are *normal*, *pre*, *nowrap*.

- The **text-shadow** property is used to set the text shadow around a text.

Example –

```
body {  
  color: blue;  
}
```

```
h1 {  
  background-color: black;  
  color: white;  
}
```

```
h1 {  
  text-align: center; | left | right | justify  
}
```

```
p {  
  direction: rtl;  
}
```

```
h1 {  
  text-shadow: 2px 2px red;  
}
```

Text shadow effect!

```
h1 {  
  text-shadow: 2px 2px 5px red;  
}
```

Text shadow effect!

5.4.3. Margin, Padding and Border Styles and Attributes

CSS Margins

https://www.tutorialspoint.com/css/css_margins.htm

The CSS `margin` properties are used to create space around elements, outside of any defined borders.

With CSS, you have full control over the margins. There are properties for setting the margin for each side of an element (top, right, bottom, and left).

Syntax –

`margin : Size;`

Margin properties can have the following values:

- Length in cm, px, pt, etc.
- Width % of the element.
- Margin calculated by the browser: `auto`.

We have the following properties to set an element margin.

- The **`margin`** specifies a shorthand property for setting the margin properties in one declaration.
- The **`margin-bottom`** specifies the bottom margin of an element.
- The **`margin-top`** specifies the top margin of an element.
- The **`margin-left`** specifies the left margin of an element.
- The **`margin-right`** specifies the right margin of an element.

If the `margin` property has one value:

- **`margin: 25px;`**
 - all four margins are 25px

If the `margin` property has two values:

- **margin: 25px 50px;**
 - top and bottom margins are 25px
 - right and left margins are 50px

If the **margin** property has three values:

- **margin: 25px 50px 75px;**
 - top margin is 25px
 - right and left margins are 50px
 - bottom margin is 75px

If the **margin** property has four values:

- **margin: 25px 50px 75px 100px;**
 - top margin is 25px
 - right margin is 50px
 - bottom margin is 75px
 - left margin is 100px

Example –

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
  background-color: lightblue;
}
</style>
</head>
<body>

<h2>Using individual margin properties</h2>

<div>This div element has a top margin of 100px, a right margin of 150px, a
bottom margin of 100px, and a left margin of 80px.</div>

</body>
</html>
```

O/P

This div element has a top margin of 100px, a right margin of 150px, a bottom margin of 100px, and a left margin of 80px.

CSS Padding -

The CSS **padding** properties are used to generate space around an element's content, inside of any defined borders. - The *padding* property allows you to specify how much space should appear between the content of an element and its border. It is important to add border properties to implement padding properties.

With CSS, you have full control over the padding. There are properties for setting the padding for each side of an element (top, right, bottom, and left).

Syntax:

```
padding: size;
```

Padding properties can have the following values:

- Length in cm, px, pt, etc.
- Width % of the element.

The following CSS properties can be used to control lists. You can also set different values for the padding on each side of the box using the following properties –

- The **padding-bottom** specifies the bottom padding of an element.
- The **padding-top** specifies the top padding of an element.
- The **padding-left** specifies the left padding of an element.
- The **padding-right** specifies the right padding of an element.
- The **padding** serves as shorthand for the preceding properties.

Example –

```

<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  background-color: lightblue;
  padding-top: 50px;
  padding-right: 30px;
  padding-bottom: 50px;
  padding-left: 80px;
}
</style>
</head>
<body>

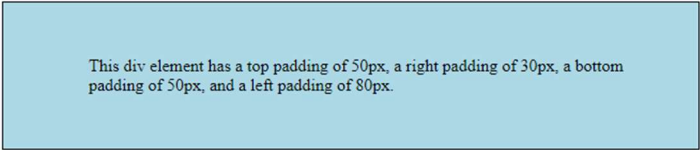
<h2>Using individual padding properties</h2>

<div>This div element has a top padding of 50px, a right padding of 30px, a
bottom padding of 50px, and a left padding of 80px.</div>

</body>
</html>

```

O/P



This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px.

If the **padding** property has four values:

- **padding: 25px 50px 75px 100px;**
 - top padding is 25px
 - right padding is 50px
 - bottom padding is 75px
 - left padding is 100px

If the **padding** property has three values:

- **padding: 25px 50px 75px;**
 - top padding is 25px
 - right and left paddings are 50px
 - bottom padding is 75px

If the **padding** property has two values:

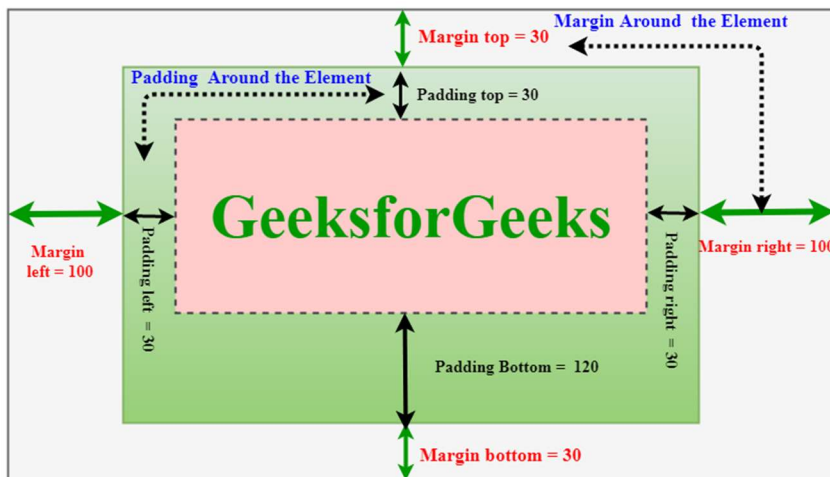
- **padding: 25px 50px;**
 - top and bottom paddings are 25px
 - right and left paddings are 50px

If the **padding** property has one value:

- **padding: 25px;**
 - all four paddings are 25px

Assignment –

Difference between margin and padding.



CSS - Borders

The **border** properties allow you to specify how the border of the box representing an element should look. There are three properties of a border you can change –

- The **border-color** specifies the color of a border.
- The **border-style** specifies whether a border should be solid, dashed line, double line, or one of the other possible values.
- The **border-width** specifies the width of a border.
- The **border-radius** property is used to add rounded borders to an element:

The **border** property is a shorthand property for the following individual border properties:

- **border-width**
- **border-style** (required)

- `border-color`

Example

```
p {
  border: 5px solid red;
}
```

O/P



The border-color Property

The border-color property allows you to change the color of the border surrounding an element. You can individually change the color of the bottom, left, top and right sides of an element's border using the properties –

- **border-bottom-color** changes the color of bottom border.
- **border-top-color** changes the color of top border.
- **border-left-color** changes the color of left border.
- **border-right-color** changes the color of right border.

Ex –

```
p.example1 {
  border: 1px solid;
  border-bottom-color: #009900; /* Green */
  border-top-color: #FF0000; /* Red */
  border-left-color: #330000; /* Black */
  border-right-color: #0000CC; /* Blue */
}
```

This example is showing all borders in different colors.

The border-style Property

The border-style property allows you to select one of the following styles of border –

- **none** – No border. (Equivalent of border-width:0;)
- **solid** – Border is a single solid line.
- **dotted** – Border is a series of dots.
- **dashed** – Border is a series of short lines.
- **double** – Border is two solid lines.
- **groove** – Border looks as though it is carved into the page.
- **ridge** – Border looks the opposite of groove.
- **inset** – Border makes the box look like it is embedded in the page.
- **outset** – Border makes the box look like it is coming out of the canvas.
- **hidden** – Same as none, except in terms of border-conflict resolution for table elements.

You can individually change the style of the bottom, left, top, and right borders of an element using the following properties –

- **border-bottom-style** changes the style of bottom border.
- **border-top-style** changes the style of top border.
- **border-left-style** changes the style of left border.
- **border-right-style** changes the style of right border.

Ex –

```
<p style = "border-width:4px;  
border-top-style:solid;  
border-bottom-style:dashed;  
border-left-style:groove;  
border-right-style:double;">  
This is a a border with four different styles.  
</p>
```



The border-width Property

The border-width property allows you to set the width of an element borders. The value of this property could be either a length in px, pt or cm or it should be set to *thin*, *medium* or *thick*.

You can individually change the width of the bottom, top, left, and right borders of an element using the following properties –

- **border-bottom-width** changes the width of bottom border.
- **border-top-width** changes the width of top border.
- **border-left-width** changes the width of left border.
- **border-right-width** changes the width of right border.

Ex –

```
<p style = "border-width:4px; border-style:solid;">  
This is a solid border whose width is 4px.  
</p>
```



```
<p style = "border-bottom-width:4px;border-top-width:10px;  
border-left-width: 2px;border-right-width:15px;border-style:solid;">  
This is a a border with four different width.  
</p>
```



Example –

```

<!DOCTYPE html>
<html>
  <head>
    <style>
      h1{
        border-style:solid;
        text-align:center;
        background:green;
        border-radius:20px;
      }
    </style>
  </head>
  <body>
    <h1>GeeksforGeeks</h1>
  </body>
</html>

```

O/P



5.4.4. List Styles and Table Layouts

CSS Lists –

https://www.tutorialspoint.com/css/css_lists.htm

Lists are very helpful in conveying a set of either numbered or bullet points. This chapter teaches you how to control list type, position, style, etc., using CSS.

We have the following five CSS properties, which can be used to control lists –

- The **list-style-type** allows you to control the shape or appearance of the marker.
- The **list-style-position** specifies whether a long point that wraps to a second line should align with the first line or start underneath the start of the marker.
- The **list-style-image** specifies an image for the marker rather than a bullet point or number.
- The **list-style** serves as shorthand for the preceding properties.

The list-style Property

The *list-style* allows you to specify all the list properties into a single expression. These properties can appear in any order.

Ex –

```
<html>
<head>
</head>

<body>
  <ul style = "list-style: inside square;">
    <li>Maths</li>
    <li>Social Science</li>
    <li>Physics</li>
  </ul>

  <ol style = "list-style: outside upper-alpha;">
    <li>Maths</li>
    <li>Social Science</li>
    <li>Physics</li>
  </ol>
</body>
</html>
```

- Maths
- Social Science
- Physics

A. Maths
B. Social Science
C. Physics

The list-style-type Property

The *list-style-type* property allows you to control the shape or style of bullet point (also known as a marker) in the case of unordered lists and the style of numbering characters in ordered lists.

Here are the values which can be used for an unordered list –

Sr.No.	Value & Description
1	none NA
2	disc (default)

	A filled-in circle
3	circle An empty circle
4	square A filled-in square

Here are the values, which can be used for an ordered list –

Value	Description	Example
decimal	Number	1,2,3,4,5
decimal-leading-zero	0 before the number	01, 02, 03, 04, 05
lower-alpha	Lowercase alphanumeric characters	a, b, c, d, e
upper-alpha	Uppercase alphanumeric characters	A, B, C, D, E
lower-roman	Lowercase Roman numerals	i, ii, iii, iv, v
upper-roman	Uppercase Roman numerals	I, II, III, IV, V
lower-greek	The marker is lower-greek	alpha, beta, gamma
lower-latin	The marker is lower-latin	a, b, c, d, e
upper-latin	The marker is upper-latin	A, B, C, D, E
hebrew	The marker is traditional Hebrew numbering	
armenian	The marker is traditional Armenian numbering	
georgian	The marker is traditional Georgian numbering	
cjk-ideographic	The marker is plain ideographic numbers	
hiragana	The marker is hiragana	a, i, u, e, o, ka, ki
katakana	The marker is katakana	A, I, U, E, O, KA, KI
hiragana-iroha	The marker is hiragana-iroha	i, ro, ha, ni, ho, he, to

katakana-iroha	The marker is katakana-iroha	I, RO, HA, NI, HO, HE, TO
----------------	------------------------------	---------------------------

Example –

```
<ul style = "list-style-type:circle;">
  <li>Maths</li>
  <li>Social Science</li>
  <li>Physics</li>
</ul>
```

- Maths
- Social Science
- Physics

The list-style-position Property

The *list-style-position* property indicates whether the marker should appear inside or outside of the box containing the bullet points. It can have one the two values –

Sr.No.	Value & Description
1	none NA
2	inside If the text goes onto a second line, the text will wrap underneath the marker. It will also appear indented to where the text would have started if the list had a value of outside.
3	outside If the text goes onto a second line, the text will be aligned with the start of the first line (to the right of the bullet).

Ex –

```
<ul style = "list-style-type:circle; list-style-position:outside;">
  <li>Maths</li>
  <li>Social Science</li>
  <li>Physics</li>
</ul>
```

```
<ul style = "list-style-type:square;list-style-position:inside;">
  <li>Maths</li>
  <li>Social Science</li>
  <li>Physics</li>
</ul>
```

- Maths
 - Social Science
 - Physics
-
- Maths
 - Social Science
 - Physics

The list-style-image Property

The *list-style-image* allows you to specify an image so that you can use your own bullet style. The syntax is similar to the background-image property with the letters url starting the value of the property followed by the URL in brackets. If it does not find the given image then default bullets are used.

Ex –

```
<ul>
  <li style = "list-style-image: url(/images/bullet.gif);">Maths</li>
  <li>Social Science</li>
  <li>Physics</li>
</ul>
```

- Maths
- Social Science
- Physics

CSS Tables –

https://www.tutorialspoint.com/css/css_tables.htm

<https://www.geeksforgeeks.org/css-tables/>

https://www.w3schools.com/css/css_table.asp

You can set following properties of a table –

- The **border-collapse** specifies whether the browser should control the appearance of the adjacent borders that touch each other or whether each cell should maintain its style.
border-collapse: collapse/separate;

- The **border-spacing** specifies the width that should appear between table cells.
- The **caption-side** captions are presented in the <caption> element. By default, these are rendered above the table in the document. You use the *caption-side* property to control the placement of the table caption.
caption-side: top/bottom;
- The **empty-cells** specifies whether the border should be shown if a cell is empty.
empty-cells: show/hide;
- The **table-layout** allows browsers to speed up layout of a table by using the first width properties it comes across for the rest of a column rather than having to load the whole table before rendering it.
table-layout: auto/fixed;

```
table, th, td {
    border: 1px solid;
}

table {
    width: 100%;
}

table {
    border-collapse: collapse;
}

th {
    height: 70px;
}

td {
    text-align: center;
}

th, td {
    padding: 15px;
    text-align: left;
}
```

5.4.5. Additional Features - Grouping Style Sheets, Assigning Classes, and Span

Grouping Style Sheets

Grouping Selectors

https://docstore.mik.ua/orelly/web2/css/ch02_02.htm

The grouping selector selects all the HTML elements with the same style definitions.

Look at the following CSS code (the h1, h2, and p elements have the same style definitions):

```
h1 {  
  text-align: center;  
  color: red;  
}  
  
h2 {  
  text-align: center;  
  color: red;  
}  
  
p {  
  text-align: center;  
  color: red;  
}
```

It will be better to group the selectors, to minimize the code.

To group selectors, separate each selector with a comma.

Example

In this example we have grouped the selectors from the code above:

```
h1, h2, p {  
  text-align: center;  
  color: red;  
}
```

Multiple Style Sheets

If some properties have been defined for the same selector (element) in different style sheets, the value from the last read style sheet will be used.

Assume that an **external style sheet** has the following style for the <h1> element:

```
h1 {  
  color: navy;  
}
```

Then, assume that an **internal style sheet** also has the following style for the <h1> element:

```
h1 {  
  color: orange;  
}
```

Example

If the internal style is defined **after** the link to the external style sheet, the <h1> elements will be "orange":

```
<head>  
<link rel="stylesheet" type="text/css" href="mystyle.css">  
<style>  
h1 {  
  color: orange;  
}  
</style>  
</head>
```

Example

However, if the internal style is defined **before** the link to the external style sheet, the <h1> elements will be "navy":

```
<head>  
<style>  
h1 {  
  color: orange;  
}  
</style>  
<link rel="stylesheet" type="text/css" href="mystyle.css">  
</head>
```

Assigning Classes, and Span

The HTML `class` attribute is used to specify a class for an HTML element.

Multiple HTML elements can share the same class.

The `class` attribute is often used to point to a class name in a style sheet. It can also be used by a JavaScript to access and manipulate elements with the specific class name.

In the following example we have three `<div>` elements with a `class` attribute with the value of "city". All of the three `<div>` elements will be styled equally according to the `.city` style definition in the head section:

Example

```
<!DOCTYPE html>
<html>
<head>
<style>
.city {
  background-color: tomato;
  color: white;
  border: 2px solid black;
  margin: 20px;
  padding: 20px;
}
</style>
</head>
<body>

<div class="city">
  <h2>London</h2>
  <p>London is the capital of England.</p>
</div>

<div class="city">
  <h2>Paris</h2>
  <p>Paris is the capital of France.</p>
</div>

</body>
</html>
```

O/P



span tag -

The `<span>` tag is an inline container used to mark up a part of a text, or a part of a document.

The `<span>` tag is easily styled by CSS or manipulated with JavaScript using the class or id attribute.

The `<span>` tag is much like the `<div>` element, but `<div>` is a block-level element and `<span>` is an inline element.

Ex –

```
<p>My mother has <span style="color:blue">blue</span> eyes.</p>
```

My mother has blue eyes

Example –

```
<!DOCTYPE html>
<html>
<head>
<style>
.note {
  font-size: 120%;
  color: red;
}
</style>
</head>
<body>

<h1>My <span class="note">Important</span> Heading</h1>
<p>This is some <span class="note">important</span> text.</p>

</body>
</html>
```

My Important Heading

This is some important text.

5.4.6. DIV Tag

The `<div>` tag defines a division or a section in an HTML document.

The `<div>` tag is used as a container for HTML elements - which is then styled with CSS or manipulated with JavaScript.

The `<div>` tag is easily styled by using the class or id attribute.

Any sort of content can be put inside the `<div>` tag!

Note: By default, browsers always place a line break before and after the `<div>` element.

Example

A `<div>` section in a document that is styled with CSS:

```
<html>
<head>
<style>
.myDiv {
  border: 5px outset red;
  background-color: lightblue;
  text-align: center;
}
</style>
</head>
<body>

<div class="myDiv">
  <h2>This is a heading in a div element</h2>
  <p>This is some text in a div element.</p>
</div>

</body>
</html>
```

O/P

The div element



This is some text outside the div element.

5.4.7. Responsive Web Design

https://www.w3schools.com/css/css_rwd_intro.asp

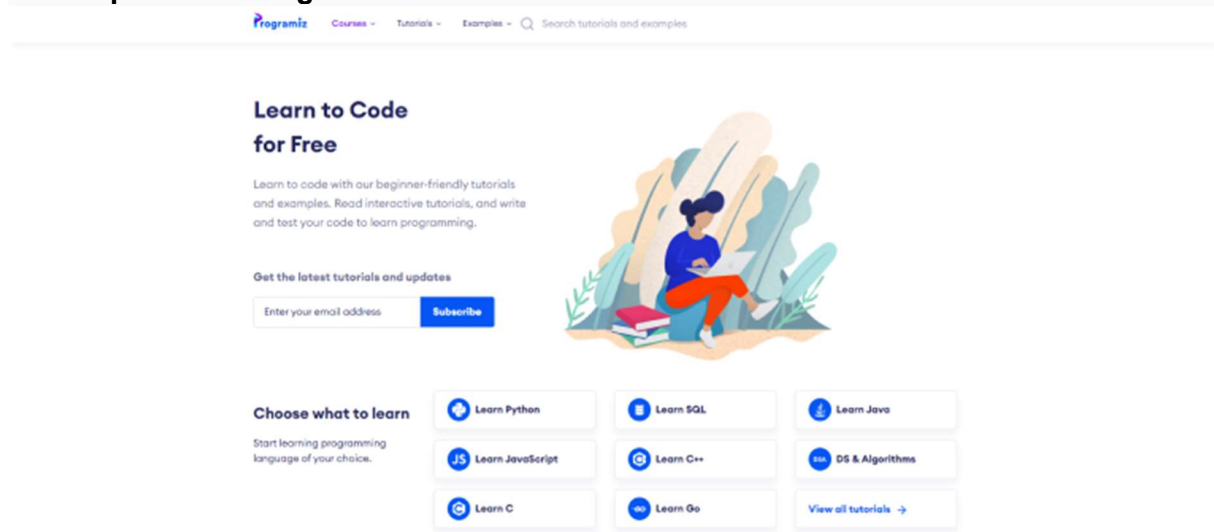
https://www.w3schools.com/html/html_responsive.asp

Responsive web design is a web design approach to make web pages render well on all devices with different screen sizes and viewports.

With the increase of devices of various sizes, it is important to create webpages that can adapt to different device sizes.

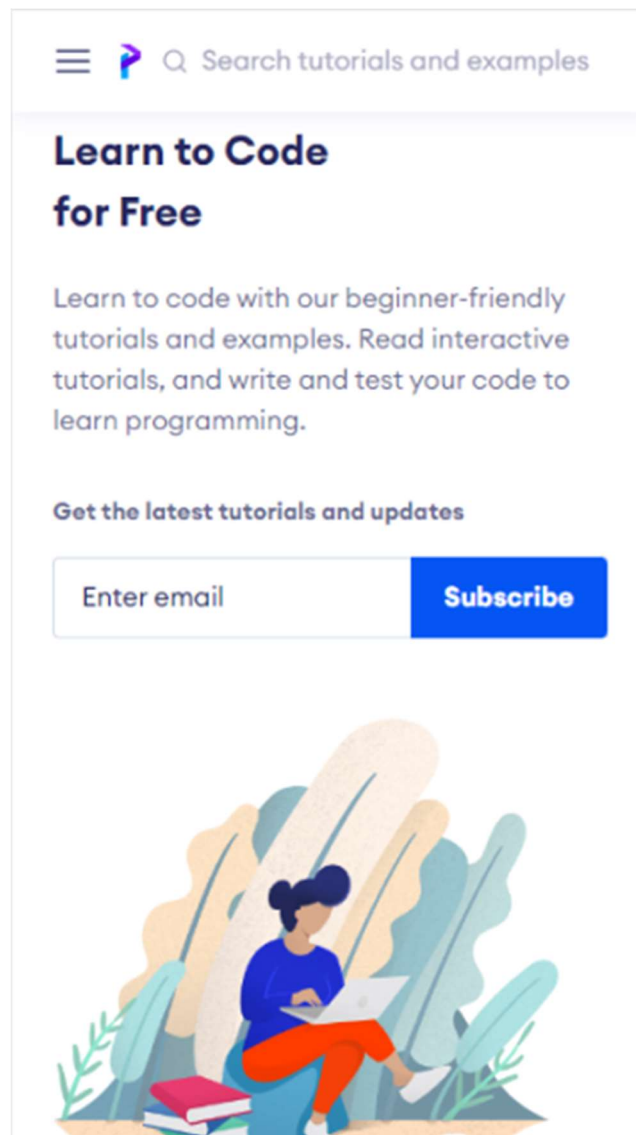
CSS media queries and other techniques allow the creation of web pages that can automatically adjust to various changes in the screen size. For example,

Desktop View of Programiz Website



Programiz Website in Desktop View

Mobile View of Programiz Website:



Programiz Website in Mobile View

Here, you can see that the design fits both mobile and desktop design.

Responsive web design makes your web page look good on all devices.

Responsive web design uses only HTML and CSS.